WHICH NOZZLE SIZE?

Choose based on the job. If you are not sure, use 0.4 mm.

SIZE	USE IT WHEN...	TRADEOFF
0.2 mm	**Tiny detail matters.** Miniatures, fine lettering, small features.	Slow. Clogs more easily. Not the default.
0.4 mm	**Almost everything.** General parts, prototypes, classroom projects.	Best all-around balance.
0.6 mm	**You want faster or tougher.** Functional parts, larger objects, thicker walls.	Less fine detail; noticeably faster.
0.8 mm	**The object is big and detail is secondary.** Jigs, fixtures, rough prototypes.	Very fast, thick lines, coarse detail.

DEFAULT: 0.4 mm

Rule of thumb: smaller nozzle = finer detail and slower printing. Larger nozzle = thicker lines, faster printing, and easier high-flow printing.

WHICH NOZZLE MATERIAL?

The filament decides whether you need a wear-resistant nozzle.

NOZZLE	USE FOR	DO NOT CHOOSE IT JUST BECAUSE...
Brass	**Standard non-abrasive filament:** PLA, PETG, ABS, ASA, TPU and similar basic materials.	It is cheap. Abrasive additives can wear it out.
Stainless steel	**When a specific application/profile calls for it.** Often chosen where material/contact requirements matter.	You think it is automatically an upgrade. It is not the normal answer for abrasives.
Hardened steel	**Abrasive filament:** carbon-fiber, glass-fiber, glow, metal/wood/mineral-filled and similar blends.	You are printing ordinary PLA. Use it when wear resistance is actually needed.

ABRASIVE FILAMENT? -> HARDENED NOZZLE

Important: nozzle material is only one compatibility limit. Abrasive or high-temperature materials can also require hardened gears, a suitable hotend, dry filament, or a different printer profile.

WHICH FILAMENT?

A practical first-choice chart - not a catalog of every printable polymer.

MATERIAL	DEFINITELY USE WHEN...	MAYBE USE WHEN...	AVOID WHEN...
PLA	Easy prototypes, models, learning, decorative parts.	Light-duty functional parts.	High heat, prolonged outdoor exposure, or flexibility is required.
PETG	You need more toughness, moisture resistance, or general utility than PLA.	Outdoor/functional parts.	You need maximum stiffness, very crisp detail, or high-temperature performance.
TPU	The part must flex: bumpers, grips, feet, gaskets.	Impact-resistant flexible parts.	You need a rigid, dimensionally stable part.
ABS	Tough, heat-resistant functional parts and you have controlled enclosure/ventilation.	Parts you may solvent-smooth or machine.	You want the easiest classroom/general-purpose print.
ASA	Outdoor parts needing UV/weather resistance.	ABS-like functional work.	You do not have appropriate enclosure/ventilation.
PA / Nylon	Mechanical parts needing toughness, wear resistance, or fatigue resistance.	Engineering parts where some flex is useful.	You cannot keep the filament dry or need easy dimensional accuracy.
PC	High-strength, higher-temperature engineering parts on a capable printer.	Very demanding functional parts.	A simpler material will do the job.

START WITH PLA. MOVE TO ANOTHER MATERIAL BECAUSE THE JOB REQUIRES IT.

Filled versions (CF/GF/glow/etc.) can change nozzle and printer requirements. Always use the printer manufacturer's material profile and compatibility guidance.

ENCLOSURE + BED: QUICK DECISION

An enclosure is not automatically better. PLA can print worse if the chamber gets too hot.

FILAMENT	ENCLOSURE?	BED / ADHESION STARTING POINT	STATION
PLA	**NO** Vent/open an enclosed printer if heat builds up.	Clean appropriate build plate. Usually no extra adhesive needed.	Open / general
PETG	Usually optional.	Clean compatible plate. On some smooth surfaces, a release layer may be recommended.	General
TPU	Usually no.	Clean compatible plate; follow surface-specific release guidance.	Flexible-material station
ABS	**YES**	Warm, stable bed/chamber; use the manufacturer's plate/profile guidance.	Enclosed engineering
ASA	**YES**	Warm, stable bed/chamber; use the manufacturer's plate/profile guidance.	Enclosed engineering
PA / Nylon	**PREFER YES** Dry filament is critical.	Dry first. Adhesion method varies substantially by PA formulation and plate.	Engineering / dry
PC	**YES**	High bed/chamber temperatures; adhesion/release layer may be specified by the plate maker.	High-temp engineering

BED RULE: CLEAN FIRST. THEN FOLLOW THE EXACT BUILD-PLATE + FILAMENT PROFILE.

Do not make 'glue stick = adhesion' a universal rule. On some plate/material combinations glue is used partly as a release layer so the print does not bond too aggressively.

Brain Gym fleet logic: open/general bank for PLA-PETG-TPU; enclosed engineering bank for ABS-ASA-PA-PC; specialty/high-end machine only when the job actually demands it.